

06 VM Instance

Create VM Instance

1. Enable Google Compute Engine API.
2. Navigate to: Virtual Machines -> VM Instances -> Create Instance -> New VM Instance -> Provide required details
3. Fill in the following fields:
 - **Name:** rev-vm
 - **Labels:** environment: dev
 - **Region:** us-central1
 - **Zone:** us-central1-a
4. Configure the machine settings:
 - **Machine Configuration:**
 - **Series:** E2
 - **Machine Type:** e2-micro
5. Set the availability policies:
 - **VM Provisioning Model:** Standard
6. Configure additional settings:
 - **Display Device:** checked
 - **Confidential VM Service:** unchecked (leave as default)
 - **Container:** unchecked (leave as default)
 - **Boot Disk:** Leave to defaults
7. Set identity and API access:
 - **Service Account:** Compute Engine default Service Account
 - **Access Scopes:** Allow default Access
8. Configure firewall settings:
 - **Allow HTTP Traffic**

The screenshot shows the 'Create an instance' wizard in Google Cloud. The 'Networking' tab is active. Under 'Firewall', the checkboxes for 'Allow HTTP traffic' and 'Allow HTTPS traffic' are checked. The 'Network tags' section shows 'http-server' and 'https-server' tags. The 'Monthly estimate' on the right shows a total of \$25.46.

```

sudo apt install -y telnet
sudo apt install -y nginx
sudo systemctl enable nginx
sudo chmod -R 755 /var/www/html
HOSTNAME=$(hostname)
sudo echo "<!DOCTYPE html>

```

```

<html>
<head>
  <title>Sample</title>
  <style>
    body {
      background-color: #1c1c1c;
      color: #ffffff;
      font-family: Arial, sans-serif;
      text-align: center;
      padding: 50px;
    }
    h1 {
      color: #e0e0e0;
    }
    p {
      font-size: 18px;
      margin: 10px 0;
    }
    .heart {
      color: #ff4081;
    }
  </style>
</head>
<body>
  <h1>Welcome to the cloud!</h1>
  <p><strong>VM Hostname:</strong> $HOSTNAME</p>
  <p><strong>VM IP Address:</strong> $(hostname -I)</p>
  <p>Google Cloud Platform - Demo 1</p>
  <h3>Made with <span class='heart'>&#10084;</span> !!!</h3>
</body>
</html>" | sudo tee /var/www/html/index.html

```

Upload and Run the webserver install script

```

# Run the commands
hostname
ip -s addr
Observation:
1. make a note of IP address and hostname

# Upload file
Upload file webserver-install.sh

# Give Permissions
chmod 755 webserver-install.sh

# Install Webserver using script
./webserver-install.sh

# Verify Files
cd /var/www/html
ls
cd app1
ls

```

Access Webserver Pages

```
# Access Webserver Pages
```

```
http://<external-ip-of-vm>
```

VM Actions

1. Start
2. Stop
3. Suspend
4. Resume
5. Reset: Reset performs a hard reset on the instance, which wipes the memory contents of the machine and resets the virtual machine to its initial state. This can lead to filesystem corruption.
6. Delete

Start

What will we lose when we stop the VM ?

Ans. External IP will be released and during the next start different public IP will be assigned to VM.

What will VM retain when we stop the VM ?

Ans. Stopped VM retains its persistent disks, internal IPs and MAC Address.

When do you stop the VM instance ?

Ans. when we don't need that VM in running state

- To save cost
- To change VM Instance Machine Type features
- To add or remove disks attached to VM Billing
- VMs in the stopped state (TERMINATED) are not charged
- Persistent Disks attached to VM, external IP Address (static) are charged

Suspend

When a VM is suspended, its memory is moved to storage.

VM Suspend: The instance will preserve its running state, similar to closing a laptop. You'll be billed for storage of the suspended VM state and any persistent disks it uses. The instance won't be accessible while it's being suspended, which may take a few minutes.

- Suspended Instances preserve the OS Memory, device state, and application state.
- When you suspend your instance, its memory moves to storage.
- When you resume your instance, its memory moves from storage to the instance memory.
- You can only suspend an instance for up to 60 days before the VM is automatically stopped.
- You cannot suspend instances with more than 120 GB of memory.
- Most of the OSs support Suspend and Resume features. However, a few OSs don't support these features.

Use Cases

- Dev and Test environments that are not used during non-business hours and need their application state to be preserved.
- Applications that take a long time to initialize when they come online.

Pricing

- Google charges for:
 - The storage necessary for the instance memory.
 - Persistent disks attached to the VM instance.
 - Any static IPs attached to the instance.

Reset and Delete

Reset VM Instance

- Performing a reset on your VM is similar to doing a hard reset on your computer.
- Resetting a VM forcibly wipes the memory contents of the machine and resets the VM to its initial state.
- The VM does not perform a clean shutdown of the guest OS.
- Throughout this process, the VM remains in the RUNNING state.

|| Delete VM Instance

- Delete an instance to remove the instance and the associated resources from your project.

|| Step-15: Where is the Boot Disk located for the demo1-vm

- Go to Compute Engine -> Storage -> Disks -> demo1-vm
- Review
 - DETAILS Tab
 - MONITORING Tab